

Practice Projects

For practice, write a function for each of the following tasks:

1. Using the head-tail technique, create a recursive `concat()` function that is passed an array of strings and returns these strings concatenated together into a single string. For example, `concat(['Hello', 'World'])` should return `HelloWorld`.
2. Using the head-tail technique, create a recursive `product()` function that is passed an array of integers and returns the total multiplied product of them. This code will be almost identical to the `sum()` function in this chapter. However, note that the base case of an array with just one integer returns the integer, and the base case of an empty array returns 1.
3. Using the flood fill algorithm, count the number of “rooms,” or enclosed spaces, in a 2D grid. You can do this by creating nested `for` loops that call the flood fill function on each character in the grid if it is a period, in order to change the periods into hash characters. For example, the following data would result in the program finding six places in the grid with periods, meaning there are five rooms (and the space outside all the rooms).

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